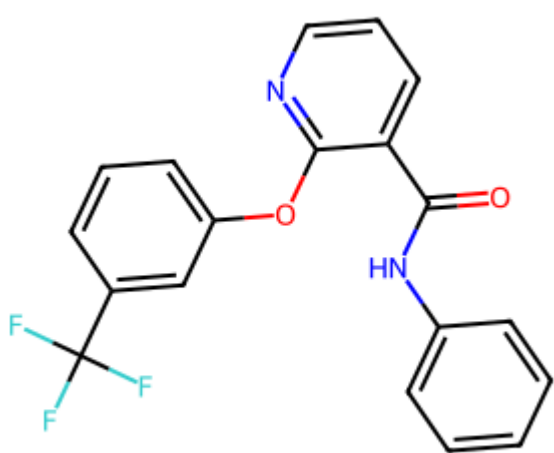
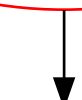


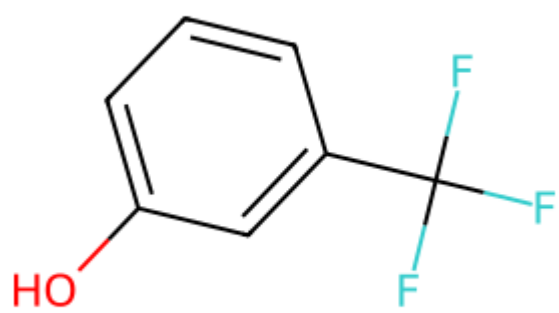


```
{'analysis_time': 0.0,
'first_solution_time': 0.238968,
'num_calls_rxn_model': 0,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 0.8568176273877603,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 0.8568176273877603,
'retro_star_value': 0.8568176273877603}
```

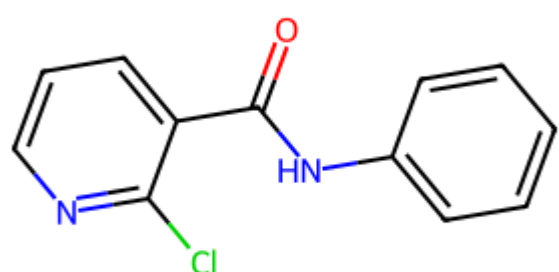


Reaction

```
{'analysis_time': 0.120389,
'first_solution_time': 0.238968,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'retro_star_min_cost': 0.8568176273877603,
'retro_star_reaction_number': 0.8568176273877603,
'retro_star_rxn_cost': 0.5485579437419287,
'retro_star_value': 0.8568176273877603}
```



```
{'analysis_time': 0.120389,
'first_solution_time': 0.120389,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 0.8568176273877603}
```

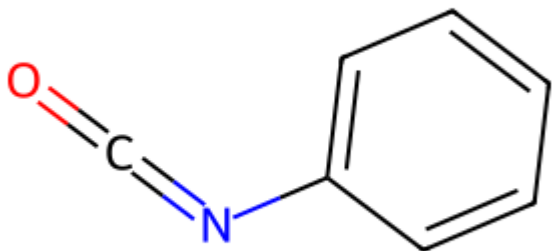


```
{'analysis_time': 0.120389,
'first_solution_time': 0.238968,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 0.3082596836458316,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 0.3082596836458316,
'retro_star_value': 0.8568176273877603}
```

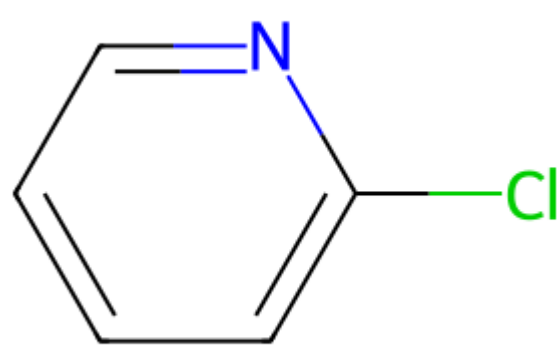


Reaction

```
{'analysis_time': 0.238968,
'first_solution_time': 0.238968,
'num_calls_rxn_model': 2,
'num_calls_value_function': 29,
'retro_star_min_cost': 11.512925464970229,
'retro_star_reaction_number': 11.512925464970229,
'retro_star_rxn_cost': 11.512925464970229,
'retro_star_value': 12.061483408712157}
```



```
{'analysis_time': 0.238968,
'first_solution_time': 0.238968,
'num_calls_rxn_model': 2,
'num_calls_value_function': 29,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 12.061483408712157}
```



```
{'analysis_time': 0.238968,
'first_solution_time': 0.238968,
'num_calls_rxn_model': 2,
'num_calls_value_function': 29,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 12.061483408712157}
```